

Movement therapy for breast cancer survivors

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Introduction

Breast cancer is the second most common type of cancer in the world and by far the most prevalent in women, with over two million new cases reported each year. It is estimated that 627,000 women died from breast cancer in 2018, that is, ~15% of all cancer deaths among women (World Health Organization, 2018). Fortunately, due to early detection and advancements in screening, many patients diagnosed will survive and go on to live out their normal life expectancy. Despite this good news, cancer treatments take an enormous toll: in just 1 year of treatment, the body can age a decade. Between the inherent weight gain, muscle atrophy and premature bone loss, women are left emotionally frail and physically weak, often challenged by normal day-to-day activities (Helms et al., 2008).

Treatments and side effects

Breast cancer treatments aim both to rid the body of disease and to minimize the possibility of the cancer returning. They are either local or systemic, with patients receiving either one form or a combination, based on individual needs. Local treatments are used to remove or destroy damaged tissue at and around the cancer site; they include surgery such as lumpectomy, mastectomy, breast-conserving surgery and axillary (underarm) lymph node removal. Systemic treatments are used to destroy or control cancer cells throughout the entire body rather than at a specific site; these methods include chemotherapy,

hormone therapy and biological therapy. Side effects to breast cancer treatments—such as pain, temporary swelling, tenderness, infection, increased stiffness in joints, “hardness” due to scar tissue that forms at the surgical site (Fourie, 2008), and very high levels of fatigue from surgery, radiation and chemotherapy—impact a patient’s ability and/or desire to exercise. Weight gain and low self-esteem are also common challenges.

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There is a growing body of exercise-oncology research that strongly suggests exercise can greatly reduce the risk of breast cancer recurrence as well as enhance functionality and quality of life (Kaelin et al., 2006; Mutrie et al., 2007; Sprague et al., 2007). It has recently been suggested that physical activity above all other possible lifestyle modifications, including weight management, diet, cessation of tobacco and reduction in alcohol consumption, may have the most robust effect on reducing breast cancer recurrence (Dieli-Conwright & Orozco, 2015). The challenge for the newly diagnosed patient is in selecting which exercise is most appropriate for them, especially if they are not currently engaged in regular physical activity. Often they look to their medical teams for advice and direction. Those who work with breast cancer, or any types of cancer patients, should be well versed in the type of cancer, the treatments administered and their associated side effects. Treatments for

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breast cancer patients differ from individual to individual due to many factors: age, extent of disease, co-morbid state and current physical condition. These factors, along with the current stage of treatment, type of medication(s) and potential side effects with exercise, should be taken into account and help guide the movement practitioner. Appropriately directed movement therapy is a critical component in the breast cancer patient's journey to recovery.

Movement therapy aims to strengthen the body-mind connection, which is essential through all phases of healing. It can provide a gentle restorative approach to exercise and is perfectly suited for women as they recover and work to rebuild their bodies both pre- and post-treatment (the benefits, most likely, also extend to the very small minority of breast cancer patients who are men). The goal of this therapeutic program is to restore the health of the damaged or compromised myofascial continuities, facilitate the slide and glide of tissues between adjacent structures, decrease excessive scar tissue build-up, build strength of the entire kinematic chain, address postural dysfunctions, help to regain sensory awareness, and teach or restore proper breathing techniques.

Many of the breast cancer patients my team and I have worked with report feeling better post-exercise and believe that movement is the key to tolerating the side effects and symptoms from cancer-related treatments, such as chemotherapy and radiation. Cancer-related fatigue (CRF) is one of the most widespread side effects and often begins immediately upon diagnosis. Studies report a prevalence of CRF in patients of up to 96% (Stasi et al., 2003). One study investigating the effects of supervised exercise training on both cardiopulmonary function and CRF in breast cancer survivors suggests that individualized exercise at a moderate intensity maintains

or improves cardiopulmonary function with associated reductions in CRF. In addition, breast cancer survivors receiving a combination of chemotherapy and radiation following surgery appear to benefit to a greater extent as a result of an individualized exercise intervention (Hsieh et al., 2008). Along with the intense bouts of CRF, breast cancer patients experience minor to often severe postural effects from cancer-related treatments (Haddad et al., 2013). Patients who undergo multiple surgical procedures including the various types of reconstructive procedures are the most compromised physiologically.

One patient came to me complaining of hip pain. She was looking for exercises to help her with her condition. While completing a comprehensive health history screen, she revealed that she had had breast cancer years before and had had a mastectomy with a transverse rectus abdominis muscle (TRAM) flap reconstruction. Her TRAM flap reconstructive surgery involved making an incision along the lower abdomen and taking an oval section of skin, fat, blood vessels, and the rectus abdominis muscle from the lower half of the belly, and rerouting it to the chest to form a breast. The rectus abdominis muscle mobilizes the trunk into flexion by pulling the ribs and pelvis inward, but it also serves as a facilitator in upright posture and, along with the deeper abdominal muscles, acts as a shield and protects the viscera. The absence of this muscle could result in movement compensation. During evaluation we found that she had overactive hip flexors with a resulting tightness of the ipsilateral (same side) lumbar erector spinae and quadratus lumborum muscles. We concluded that her hip pain was likely due to her years of movement compensation post-breast cancer surgery and not necessarily an isolated hip joint problem. The good news is that after following a protocol similar to the one outlined in this chapter, within 3 months she was pain-free, and within 9 months her audible hip clicking with basic everyday movements was gone. She was able to resume her passion for long walks and hiking without discomfort or pain.

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Postural dysfunctions resulting from multiple life-saving surgical procedures and treatments require careful attention and specifically designed exercise regimens to help re-establish proper neuromuscular patterns and restore optimal posture.

Potential movement therapy benefits for breast cancer survivors

(N.B. The benefits listed below have not been proven and are hypothetical; more research is needed before they can be stated as absolutes).

Movement therapy

- improves shoulder girdle mechanics, specifically scapula-humeral rhythm, to aid in the breakdown of scar tissue and frozen shoulder
- improves lymphatic drainage through breathing
- restores overall range of motion, flexibility
- re-establishes proper muscular firing patterns
- increases the strength of local and global stabilizing muscles
- enhances core muscular strength and endurance
- restores upright postural alignment and balance
- improves overall sensory awareness
- assists with weight management
- increases self-confidence and overall well-being
- reduces both physiological and emotional stress
- improves appearance.

The key to the success of any exercise and movement-based therapy program is to proceed slowly and to consistently monitor how the

patient feels. Developing a continual dialogue will ensure that they are not taking on more than they can handle. Just as crucial is the practitioner's bedside manner; there is a strong emotional element when working with breast cancer survivors, and positive reinforcement from the wellness professional is vital in all phases.

Movement therapy program design: five key concerns

It is imperative that movement professionals understand that patients recovering from breast cancer differ from those healing from injury and address the physiological conditions resulting from treatment (Kaelin et al., 2006). Movement therapy practitioners should take these conditions into account when implementing exercise programs for breast cancer survivors.

Lymphedema

This condition relates specifically to axillary lymph node dissection, axillary radiation or surgical infection (swelling of the arm). The swelling occurs when the lymph channels are altered and no longer able to properly drain lymph fluid from the arm back into the body's general circulation. Patients who have developed lymphedema should follow the guidelines outlined by their medical team and may be advised to wear compression garments when exercising. Lymphedema is managed by compression garment wear, exercise and elevation (APTA, 2020).

Rotator cuff

Breast cancer surgeries affect the rotator cuff muscles, resulting in "faulty" shoulder girdle mechanics, poor posture, increased stiffness, decreased mobility and pain (Ebaugh et al., 2011). Restoring the function of these muscles will help with daily life activities that involve pulling and pushing actions.

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Sarcopenia

This condition results in a simultaneous loss of muscle and gain in fat tissue. Inactivity during treatment, chemotherapy, premature menopause and other hormonal changes brought on by breast cancer treatments may lead to sarcopenia. Sarcopenia is associated with an increased risk of overall mortality in breast cancer survivors and may be associated with breast cancer-specific mortality (Villaseñor et al., 2012).

Premature osteoporosis

As a result of both local and systemic treatment, many breast cancer survivors experience a loss of ovarian function and, consequently, a drop in estrogen levels. This can increase bone loss and lead to osteoporosis. Women who were premenopausal before their cancer treatment may go through menopause earlier than those who have not had breast cancer (National Institutes of Health Osteoporosis and Related Bone Diseases National Resource Center, 2016).

Physiological muscular imbalances

After a mastectomy, a woman may opt to have surgery to reconstruct her breast(s) using an artificial implant or her own tissue, referred to as an autologous tissue flap. Flap surgery takes tissue from the abdomen, back or buttocks and uses it to reconstruct the breast. The effect of this procedure varies significantly from patient to patient. Flap surgeries often create muscular imbalances, and compensation patterns. Most patients have no difficulty in day-to-day life activities (Cyriac et al., 2010). However, there is insufficient long-term research detailing the effects on the long-term function of the associated prime movers, such as the rectus abdominis, gluteus and latissimus dorsi. It can be assumed

that loss of prime mover function may result in a greater demand on synergistic muscles, such as the obliques, rhomboids, and mid and lower trapezius.

The relationship between cancer and fascia

While the relationship between cancer and fascia has been a relatively unexplored topic in the scientific and therapeutic worlds, researchers are starting to recognize the potential importance of fascia in cancer, especially when it comes to integrative medicine, not just drugs and pharmacology (Langevin et al., 2016). Therapeutic work to rebuild the body may involve rehabilitation of the fascia, which could have been damaged in treatment.

Reports have shown that individuals with dense breast tissue have a 4- to 6-fold increased risk of developing breast carcinomas and that a third of all breast cancer cases are attributed to breast density, making it one of the greatest risk factors for carcinoma (Provenzano et al., 2008). When the tissue underlying the tumor is stiffer, it encourages the cancer to not only grow but to spread. In addition, the cancer secretes factors that increase the stiffness of the tissue creating a vicious cycle of disease (Provenzano et al., 2008). The stiffness and architecture of the collagen in the connective tissue plays a role in cancer progression. Collagen fiber bundles with a specific structured organization around tumors, as opposed to unstructured wavy collagen fibers, have been shown to be more evident in cancer that has spread, as in metastatic breast cancer. These structured collagen fibers contribute to cancer progression and facilitate invasion out from primary tumors (Provenzano et al., 2008). While there is not much we can do

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to change one's predisposed breast tissue density, the knowledge of how tissue density impacts the disease can help direct our therapeutic approach with patients both during and post-cancer treatments. We will want to minimize the build-up of scar tissue and restore tissue mobility as quickly and as safely as possible. Recommendations in programming should include diaphragmatic breathing techniques and stretching or release methods to enhance the range of motion and improve movement quality.

Survivorship is a complex road in a breast cancer patient's journey. In addition to the enormous physical challenges, many patients experience psychological trauma resulting in depression, anxiety and fear. Dealing with a life-threatening disease and living with extended periods of stress can often lead to an increase in sympathetic nervous system activity (Won and Kim, 2016). Findings have shown that sympathetic activation may cause fascial contractions and potentially lead to an increase in fascial stiffness (Staubesand and Li, 1996; Bhowmick et al., 2009). The combination of moderate to low intensity movement therapy with integrative, contemplative practices such as yoga, meditation, Pilates and tai chi has shown improvements in sleep, anxiety, depression and quality of life for cancer patients (Buettner et al., 2006; Stan et al., 2012). Given that contemplative practices help decrease sympathetic tone, it could be speculated that it may, in turn, help decrease fascial stiffness.

Inflammation is another exercise-oncology research topic of interest. Inflammation is an extremely problematic human condition and is related to chronic musculoskeletal pain, different types of inflammatory diseases and cancer. Utilizing physically based treatments, known as mechano-therapeutics, as opposed to drugs

may play a positive role in the development and progression of cancer cells/tumors. Factors such as tissue stiffness, architecture of the tissue, and tissue forces in the matrix have a profound effect on the sustainability and progressive nature of cancers. When it comes to forces, there are many unanswered questions: How much force? How should it be applied and for how long? (Berrueta et al., 2018).

For much of the 20th century, the vast majority of scientific efforts to understand and destroy cancer have focused on the cells, but not on the environment in which they live: the microenvironment. Cells in the human body, when operating under normal conditions, have specific jobs to do. They receive instructions from their genetic material and from signals in their environment and respond by obeying these commands. Langevin (2018) states that "cancer cells are cells that have gone rogue: they march to the beat of their own drum." Under normal circumstances there is a dynamic yet stable interaction between cells and their surrounding extra cellular matrix (ECM). Cancer development results when there are alterations to those normal interactions. Cancer cells appear to behave as masters of division, multiplying out of control and not stopping when it is appropriate, gradually invading the body.

Cellular response to both chronically inflamed tissue and cancer-associated tissues is to become contractile, increasing stiffness in the ECM by generating sustained and continued tension on the ECM. Increased stiffness in the ECM is accompanied by greater interstitial pressure and lymphatic flow, which may predispose an individual to fibrosis, cancer invasion and metastasis. Mechanical factors, i.e. physical force, can be a potential target for

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therapeutic intervention. As observed in animal models, just 10 min per day of simple stretching can potentially slow down the advancement of cancer (Langevin, 2018).

Movement therapy: phases of recovery

The movement therapy exercises outlined in this chapter are a sample of movements that may prove helpful with the various phases of breast cancer healing and recovery:

(1) Pre-habilitation (prehab) is a preparatory phase that begins as early as the day of diagnosis and continues until medical treatments commence. In this phase you are able to provide movement education and to prepare the patient for surgery and cancer treatment.

(2) Post-rehabilitation – phases I and II. The rehabilitation process is broken down into two segments that begins after preliminary tissue rehabilitation has occurred and/or the medical team has cleared the patient to perform a gentle range of motion and low-load strengthening. These phases are designed to re-educate the body-mind connection and to develop a strong physical foundation. The exercises can be used throughout the course of the patient's cancer journey to restore and rebuild their bodies.

Diagnosis and prehab

Upon diagnosis, patients undergo a continuum of defensive and protective behaviors, both emotional and physical. This is the time to begin preparing the mind and body for what is to come. Because this is pre-treatment, there are not many exercise restrictions. However, there should be a significant focus on muscular strength and endurance, especially of the upper body and core. Daily movement is key to keep

both the mind stimulated and all systems of the body functioning.

For almost a decade, Hubert Godard and a team of researchers from the National Cancer Institute in Milan observed the physical and psychological damage of women with breast cancer. They conducted experiments with different groups of patients while walking and performing movements of daily life. One group was in the pre-operative phase and the other was in the post-surgical phase. In the pre-operative group, the researchers observed that as many as 50% of women had already begun to lose pendulousness or rhythmic swing quality in their affected limbs (arms) with just the news of the diagnosis, and also that the pendulousness would further decrease after surgery, in some cases as a complete arrest of the arm. In the post-surgical group, the researchers found unusual post-surgical limitations of the shoulder girdle, which could not be explained through traditional examinations. When walking, 65% of women demonstrated a loss of pendular motion in the arm on the affected side. The arm appeared anchored to the trunk, almost lifeless. Yet, when prompted to swing their arms symmetrically, they did so with no difficulty (Godard and Martino, 2001).

As witnessed in these experiments, patients diagnosed with breast cancer will begin to guard their body, subconsciously protecting the arm on the affected side, keeping it close to their side, not allowing it to hang loosely, or even swing naturally with gait. The early onset of this protective behavior may result in a physical manifestation, and greatly impact the health and function of the muscles and connective tissue, making them more rigid and stiff. Within as little as 2 to 3 weeks, immobilized rat limbs become rigid, lose elasticity and the connective tissue fibers become adhesive (Jarvinen, 2002). In my experience, the patients who enter the treatment process more aware of their bodies

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and physically stronger have an easier time dealing with the physical limitations post-surgery. And as evidenced by Godard and Martino (2001), the patients were not aware they had developed this protective mechanism. The overall body posture in women who undergo mastectomies may be negatively affected, especially for those with large, heavy breasts (Haddad et al., 2013). Restrictions in the upper limbs and difficulty with upper-body tasks leave many women feeling physically and psychologically drained: the dysfunctions may remind them of their illness and lack of potential or full capacity in life (Collins et al., 2004). For the breast cancer patient, movement awareness and education is critical in the recovery process and may help with some of the painful and debilitating postural side effects from surgical and systemic treatments (Hojan et al., 2013).

Potential benefits of prehab movement therapy

- Maintain energy
- Remain emotionally grounded
- Build stamina
- Increase circulation and overall blood flow
- Foster mind-body connection
- Strengthen areas likely compromised by local and systemic treatments

Post-rehab phase I: movement therapy for breast cancer

At the first session, ensure you review the client's overall health history as well as the types of treatments and surgeries which they may have undergone. This information is instrumental in developing an effective and safe movement therapy program. Remember that you are not

here to diagnose but to help with rehabilitation. Therefore, it is imperative that you prepare by gathering all of the facts before you begin. Keep in mind that the usual stages for tissue healing and exercise progression for a client who has had cancer differ greatly from a client who has an acute or chronic injury. Symptoms like muscle tightness and joint stiffness are present in both instances. However, while the symptomology may appear similar, you cannot treat a mastectomy or a lumpectomy as you would a rotator cuff or joint injury. Treatments such as chemotherapy and radiation coupled with surgery greatly affect the time it takes to heal and fully recover. Also, keep in mind that there are ongoing side effects that this client may experience which are unlike those an injured client may have.

Phase I exercises focus on developing the client's kinesthetic awareness and teach very basic biomechanical movement principles. These principles are foundational and can be part of the client's daily exercise regimen forever. After the initial screening, introduce simple biomechanical movement principles. The basic principles selected here are essential to both dynamic stability and mobility. These include:

- Breathing
- Pelvic placement
- Ribcage placement
- Scapula movement and stabilization
- Head and cervical placement

(Further information is available at <https://www.merrithew.com/stott-pilates-basic-principles>.)

In addition to the previously mentioned principles, listed below are five additional exercises which are appropriate for phase I.

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Focused breathing with mini-stability balls

Benefits: regulate the lymphatic system and rid the body of toxins. In addition to gravity and muscular contractions, the breath serves as the primary pump for the lymphatic system. The breath also encourages engagement of the deep core musculature: transversus abdominis, internal obliques, pelvic floor and multifidus, all of which are important for restoring posture and functional strength.

Start position: seated, kneeling or standing, holding the ball at chest or underarm. Choose ball size according to comfort level and range of motion.

Exercise and breath pattern: inhale deeply into the ball, exhale allow the ball to gently press against body. Stay for 3–5 full breaths in each position.

Beneficial cues: use the ball as a focal point to send the breath, expand the breath into the ball.

Limb bouncing – arms

Benefits: bouncing fosters rhythmic movement, spring-like, non-muscular, almost effortless action, and is intended to calm and restore.

Start position: lying supine on mat, knees bent.

(A) Arm by side with blue ball

(B) Arm in flexion with orange ball

Pre-tension arm before bouncing to feel tensional integrity in limb and to spark and sustain rhythmic bouncing.

Exercise and breath pattern: breathing is not dictated, it is natural. Stay for 20–30 s or until the movement feels like an effort.



Figure 46.1 Focused Breathing with mini stability balls

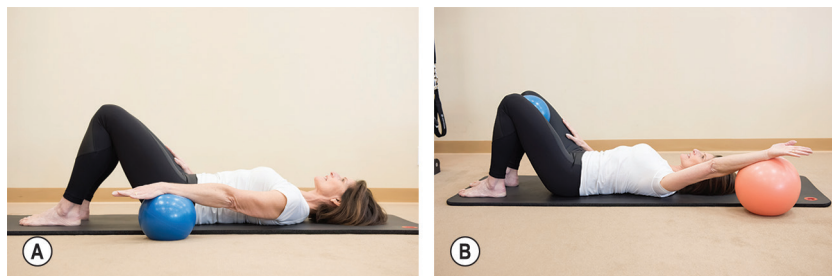


Figure 46.2 A&B Limb Bouncing – Arms

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Beneficial cues: music is beneficial to foster one's rhythmic quality. Music at ~130 beats per min is optimal with arm bouncing.

Limb bouncing – legs

Benefits: bouncing fosters rhythmic movement, spring-like, non-muscular, almost effortless action, and is intended to calm and restore.

Start position: lying supine on mat with knees bent. Pre-tension leg before bouncing to feel tensional integrity in limb and to spark and sustain rhythmic bouncing.

Exercise and breath pattern: breathing is not dictated, it is natural. Stay for 20–30 s or until the movement feels like an effort.

Beneficial cues: music is beneficial to foster one's rhythmic quality when bouncing. Music at ~120–130 beats per min is optimal with leg bouncing.

Hamstring curls with sliders

Benefits: works the posterior chain, which is critical in regaining verticality and good posture. The slider provides proprioceptive stimulus to enhance stability of core musculature

and enable an ease of glide and facilitate smooth movement.

Start position: lying supine on mat with entire spine neutral, knees bent, feet dorsiflexed with heels on the sliders.

Exercise and breath pattern: inhale, slide both feet out as far as a neutral spine can be maintained exhale. Slide both feet back in to start position. Repeat 8–12 times.

Beneficial cues: apply just enough pressure to feel the muscles activate, use the breath to help control the movement. Keep upper body relaxed and focus on keeping trunk stable.

Standing hinge and low row with resistance loop

Benefits: hinging is a primary foundational movement pattern, necessary in many functional moves of daily living. Rows strengthen the shoulder girdle and posterior chain.

Start position: standing on resistance loop, holding ends of loop, palms facing in.

Exercise and breath pattern: inhale to hinge at hip maintaining neutral spine (Figure 46.5B),



Figure 46.3 Limb Bouncing – Legs

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Figure 46.4 Hamstring Curls with Sliders



Figure 46.5 A-C Standing Hinge & Low Row with Resistance Loop

exhale, squeeze shoulder blades together and slightly down, and pull band in close to the body (Figure 46.5C). Repeat row pattern 8–12 times.

Beneficial cues: keep body from shifting back when hinging at hips, keep spine neutral throughout.

Movement therapists should remember that it is critical in phase I to avoid taxing or overworking any one area of the body. Select one or two exercises for the upper body and then one or two for the lower body. Follow the recommended repetitions but pay close attention to the ability of the client and let the weaker side determine the amount of resistance and number of repetitions.

Always support the affected limbs with pillows or cushions when necessary, and only work in a comfortable range of motion. Never work through pain.

Post-rehab phase II – movement therapy for breast cancer

The goal of the phase II exercises is to continue building upon the foundations set in the previous phase. These foundations focus on restoring joint mobility with a range of gentle motion exercises designed to break down residual scar tissue, both from surgery and various treatments. We continue our work to increase

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overall body awareness with slow, controlled and concentrated movements. Introduce the following low-load exercises to strengthen the local stabilizers, promote joint stability and enhance neuromuscular control. Use a layering effect as you add each new exercise, and keep a record of what you have added. This will help you to determine which exercises the client can withstand and which may be overly challenging. If you add too many new exercises at once then you will not know which move could be problematic.

Limb bouncing – arms

Benefits: bouncing fosters rhythmic movement, spring-like, non-muscular, almost effortless action, and is intended to calm and restore.

Start position: supine on mat, arm in flexion overhead resting on ball, palm up or in any comfortable position. Pre-tension arm before bouncing to feel tensional integrity in limb and to spark and sustain rhythmic bouncing.

Exercise and breath pattern: breathing is not dictated, it is natural. Stay for 20–30 s or until the movement feels like an effort.

Beneficial cues: music is beneficial to foster one's rhythmic quality when bouncing. Music at ~130 beats per min is optimal with arm bouncing.

Hamstring curl bridge combo

Benefits: this version adds the bridge, which will increase the workload of the posterior chain and place additional emphasis on the core.

Start position: lying supine with hips extended to a lower hover over mat, knees bent, feet dorsiflexed with heels on the sliders.

Exercise and breath pattern: inhale, slide one foot out as far as a neutral spine can be maintained. Exhale, return to start position. Repeat, alternating legs 5–8 times.

Note: to progress this exercise slide both feet out.

Beneficial cues: apply just enough pressure to feel the muscles activate, use the breath to help control the movement, keep the upper body relaxed and focus on keeping trunk stable.

Standing flexion and abduction

Benefits: increase strength in abductors of hip, improves posture and balance.



Figure 46.6 Limb Bouncing – Arms

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Figure 46.7 A-C Hamstring Curl Bridge Combo



Figure 46.8 A-C Standing Flexion & Abduction

Start position: standing sideways about one foot distance from wall. Place inside arm against wall for balance.

Exercise and breath pattern: inhale to lift inside knee to hip height, exhale press knee against wall (Figure 46.8B) and if possible bring arms to the genie position in front of body for balance (Figure 46.8C). Stay and breath for 2–3 full breaths and return to start position. Repeat three times on each side of the body.

Beneficial cues: keep tension on lateral aspect of outside leg but maintain neutral foot and knee alignment, stay tall.

Standing low row with resistance loop

Benefits: enhances slide and glide of scapula, increases strength in shoulder girdle muscles, excellent for posture. The loop anchored high will strengthen the lower trapezius muscles.

Start position: standing facing anchor of resistance loop (loop anchored high), hold loop with

hands, palms facing in, arms extended long in front of body.

Exercise and breath pattern: inhale, prepare. Exhale, squeeze shoulder blades together and slightly down, and pull band in close to body. Repeat row pattern 8–12 times.

Beneficial cues: think of the scapula sliding in and down the back.

Forearm plank scapula isolations

Benefits: the plank is the gold standard in core stability, using the ball supports the trunk and provides a good assist until strength to hold full plank is achieved.

Start position: prone with forearms on mat, orange ball placed under the pelvis and lower trunk, knees bent, toes curled under.

Exercise and breath pattern: inhale to prepare. Exhale to lift knees off mat to full plank position. Remain in plank for 3–5 full breaths breathing normally.

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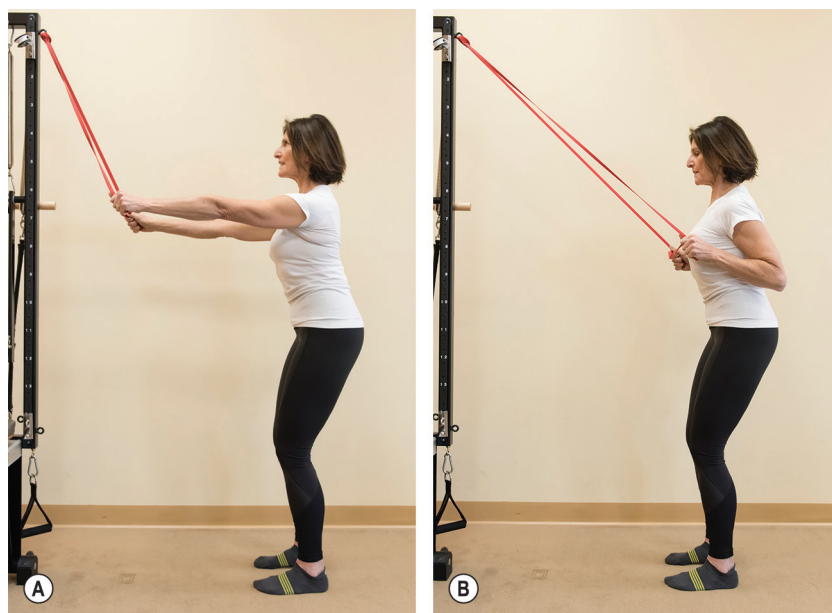


Figure 46.9 A&B Standing Low Row with resistance loop

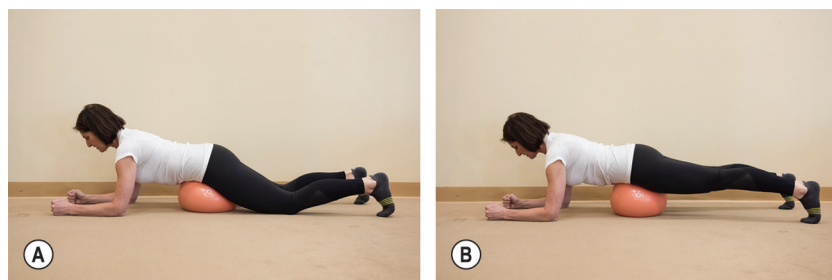


Figure 46.10 A&B Forearm Plank Scapula Isolations

Beneficial cues: keep the body taut to avoid sagging or collapse. Use the breath to maintain posture. Avoid sinking into the ball.

Be mindful of these phase II contraindications:

- Progressing too quickly
- Using too heavy a load and potentially stressing the joints
- Not adhering to a client's needs and physician recommendations
- Overworking the limbs, especially in the affected areas
- Not allowing for adequate rest and recovery between sessions

A strong component of body-mind exercise is empowering the client by increasing their kinesthetic awareness. As this happens, the client becomes more familiar with her own body from one day to the next and can recognize when things are not quite right.

By arming our clients with simple, effective exercises, we provide them with the tools they need to become self-reliant, to enjoy a higher quality of life and be more functional in everyday activities.

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